Model

```
#getting rid of the data we will not be using in the model
#loading in the necessary package
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
# Load in start stuff
library(brms) # for statistics
```

```
Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').


Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ forcats   1.0.1     ✔ readr     2.1.6
✔ ggplot2   4.0.1     ✔ stringr   1.6.0
✔ lubridate 1.9.4     ✔ tibble    3.3.1
✔ purrr     1.2.0     ✔ tidyr     1.3.2

── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

```
library(ggeffects) # for the prediction plot
```

The Model

Load Data and Create Proportions

```
library(brms)
library(dplyr)

Whale_Data_Filtered <- read.csv("Whale_Data_Filtered.csv")

eps <- 0.0001

Whale_Data_Filtered <- Whale_Data_Filtered |>
  mutate(
    Total = Resting.... + Traveling.... + Foraging.... + Exploring....,
    Resting_p   = Resting....   / Total + eps,
    Traveling_p = Traveling.... / Total + eps,
    Foraging_p  = Foraging....  / Total + eps,
    Exploring_p = Exploring.... / Total + eps
  ) |>
  mutate(
    total = Resting_p + Traveling_p + Foraging_p + Exploring_p,
    Resting_p   = Resting_p / total,
    Traveling_p = Traveling_p / total,
    Foraging_p  = Foraging_p / total,
    Exploring_p = Exploring_p / total
  )
```

Run behavior models

```
behaviors <- c("Resting_p","Traveling_p","Foraging_p","Exploring_p")

models <- lapply(behaviors, function(b){
  brm(
    as.formula(paste(b,"~ Group")),
    data = Whale_Data_Filtered,
    family = Beta(),
    chains = 4,
    cores = 4,
    iter = 4000
  )
})
```

```
Compiling Stan program...
```

```
Trying to compile a simple C file

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: ‘Apple clang version 17.0.0 (clang-1700.6.4.2)’
using SDK: ‘MacOSX26.2.sdk’
clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/Rcpp/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/unsupported"  -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/BH/include" -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/src/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppParallel/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/rstan/include" -
DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -
DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -
D_HAS_AUTO_PTR_ETC=0  -include '/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp'  -D_REENTRANT -
DRCPP_PARALLEL_USE_TBB=1   -I/opt/R/arm64/include    -fPIC  -falign-functions=64 -Wall -g
-O2  -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal
error: 'cmath' file not found
  679 | #include <cmath>
      |          ^~~~~~
1 error generated.
make: *** [foo.o] Error 1

Start sampling

Compiling Stan program...

Trying to compile a simple C file

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: ‘Apple clang version 17.0.0 (clang-1700.6.4.2)’
using SDK: ‘MacOSX26.2.sdk’
clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/Rcpp/include/"  -
```

```
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/unsupported"  -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/BH/include" -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/src/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppParallel/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/rstan/include" -
DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -
DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -
D_HAS_AUTO_PTR_ETC=0  -include '/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp'  -D_REENTRANT -
DRCPP_PARALLEL_USE_TBB=1   -I/opt/R/arm64/include    -fPIC  -falign-functions=64 -Wall -g
-O2  -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal
error: 'cmath' file not found
  679 | #include <cmath>
      |          ^~~~~~
1 error generated.
make: *** [foo.o] Error 1
00 / 4000 [ 80%]  (Sampling)
Chain 2: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 3: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 1: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.067 seconds (Warm-up)
Chain 1:                0.066 seconds (Sampling)
Chain 1:                0.133 seconds (Total)
Chain 1:
Chain 4: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 3: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.066 seconds (Warm-up)
Chain 3:                0.068 seconds (Sampling)
Chain 3:                0.134 seconds (Total)
Chain 3:
Chain 2: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.065 seconds (Warm-up)
```

```
Chain 2:                0.074 seconds (Sampling)
Chain 2:                0.139 seconds (Total)
Chain 2:
Chain 4: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.067 seconds (Warm-up)
Chain 4:                0.071 seconds (Sampling)
Chain 4:                0.138 seconds (Total)
Chain 4:

Start sampling

Compiling Stan program...

Trying to compile a simple C file

SSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS
-DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
'/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp'  -D_REENTRANT -
DRCPP_PARALLEL_USE_TBB=1   -I/opt/R/arm64/include    -fPIC  -falign-functions=64 -Wall -g
-O2  -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal
error: 'cmath' file not found
  679 | #include <cmath>
      |          ^~~~~~
1 error generated.
make: *** [foo.o] Error 1
3600 / 4000 [ 90%]  (Sampling)
Chain 1: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 2: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.068 seconds (Warm-up)
Chain 1:                0.069 seconds (Sampling)
Chain 1:                0.137 seconds (Total)
Chain 1:
Chain 4: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 3: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.068 seconds (Warm-up)
Chain 3:                0.066 seconds (Sampling)
Chain 3:                0.134 seconds (Total)
Chain 3:
```

```
Chain 2: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.069 seconds (Warm-up)
Chain 2:                0.076 seconds (Sampling)
Chain 2:                0.145 seconds (Total)
Chain 2:
Chain 4: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.069 seconds (Warm-up)
Chain 4:                0.07 seconds (Sampling)
Chain 4:                0.139 seconds (Total)
Chain 4:

Start sampling

Compiling Stan program...

Trying to compile a simple C file

rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include
'/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp'  -D_REENTRANT -
DRCPP_PARALLEL_USE_TBB=1   -I/opt/R/arm64/include    -fPIC  -falign-functions=64 -Wall -g
-O2  -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp:22:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Dense:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/Core:19:
/Library/Frameworks/R.framework/Versions/4.4-
arm64/Resources/library/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10: fatal
error: 'cmath' file not found
  679 | #include <cmath>
      |          ^~~~~~
1 error generated.
make: *** [foo.o] Error 1
00 / 4000 [ 90%]  (Sampling)
Chain 4: Iteration: 3200 / 4000 [ 80%]  (Sampling)
Chain 1: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.071 seconds (Warm-up)
Chain 1:                0.065 seconds (Sampling)
Chain 1:                0.136 seconds (Total)
Chain 1:
Chain 3: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 2: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 2:
```

```
Chain 2:  Elapsed Time: 0.068 seconds (Warm-up)
Chain 2:                0.07 seconds (Sampling)
Chain 2:                0.138 seconds (Total)
Chain 2:
Chain 4: Iteration: 3600 / 4000 [ 90%]  (Sampling)
Chain 3: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.068 seconds (Warm-up)
Chain 3:                0.073 seconds (Sampling)
Chain 3:                0.141 seconds (Total)
Chain 3:
Chain 4: Iteration: 4000 / 4000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.072 seconds (Warm-up)
Chain 4:                0.074 seconds (Sampling)
Chain 4:                0.146 seconds (Total)
Chain 4:

Start sampling
```

```r
names(models) <- behaviors
```

Extract results

```r
behavior_results <- data.frame(
  Behavior = behaviors,
  Estimate = sapply(models, function(m) fixef(m)[2,"Estimate"]),
  LowerCI  = sapply(models, function(m) fixef(m)[2,"Q2.5"]),
  UpperCI  = sapply(models, function(m) fixef(m)[2,"Q97.5"])
)

behavior_results
```

```
               Behavior     Estimate    LowerCI   UpperCI
Resting_p     Resting_p  0.002785671 -0.7344003 0.6261247
Traveling_p Traveling_p  0.005863524 -1.0129384 0.8188505
Foraging_p   Foraging_p -0.595299599 -1.8206870 0.4663888
Exploring_p Exploring_p  0.226044855 -0.2621674 0.7666207
```

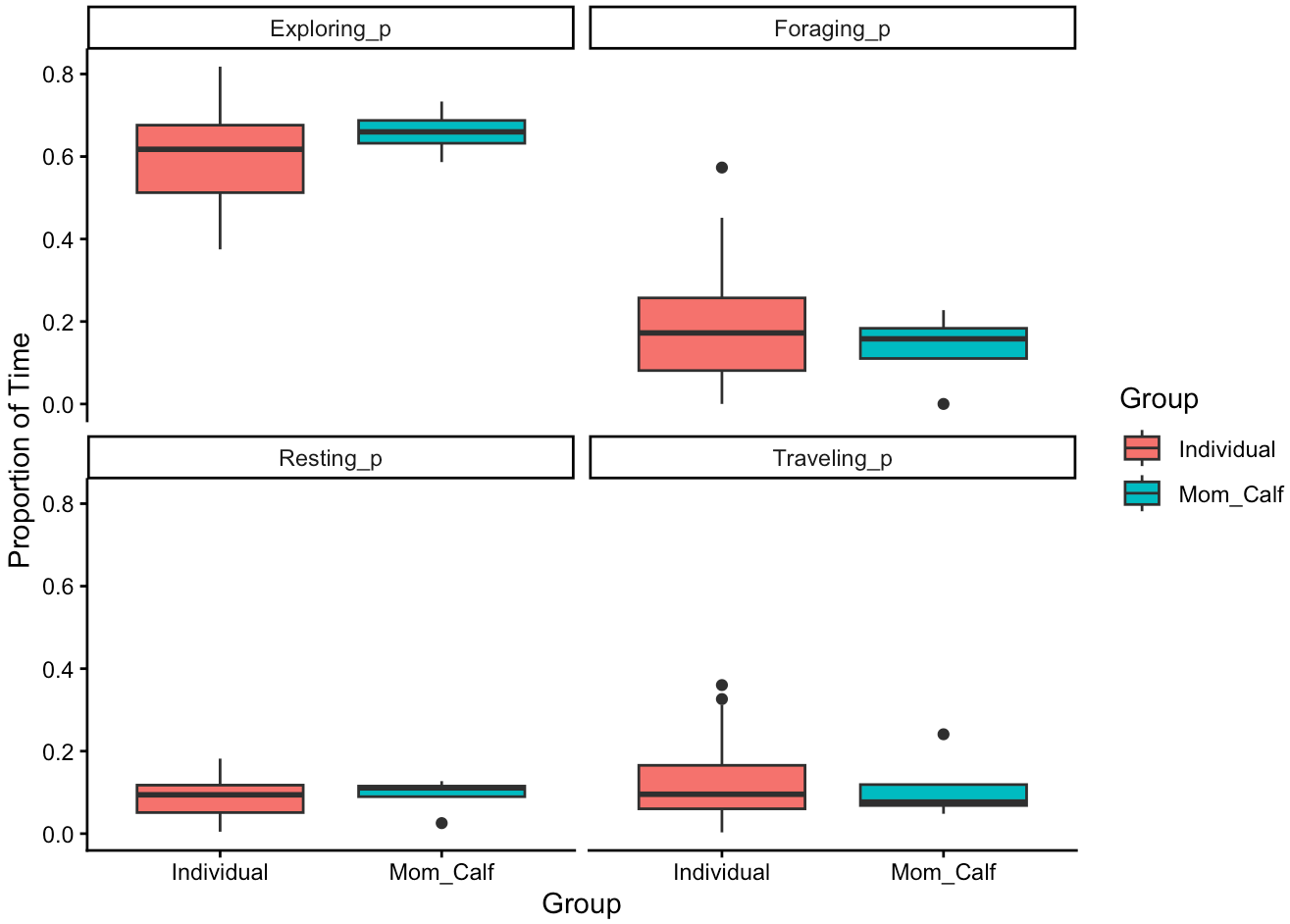
Plot behavior proportions

```r
library(tidyr)
library(ggplot2)

# ----- BOXPLOT FIGURE -----
```

```
Whale_Long <- Whale_Data_Filtered |>
  pivot_longer(
    cols = Resting_p:Exploring_p,
    names_to = "Behavior",
    values_to = "Proportion"
  )

ggplot(Whale_Long, aes(Group, Proportion, fill = Group)) +
  geom_boxplot() +
  facet_wrap(~Behavior) +
  theme_classic() +
  labs(x="Group", y="Proportion of Time")
```

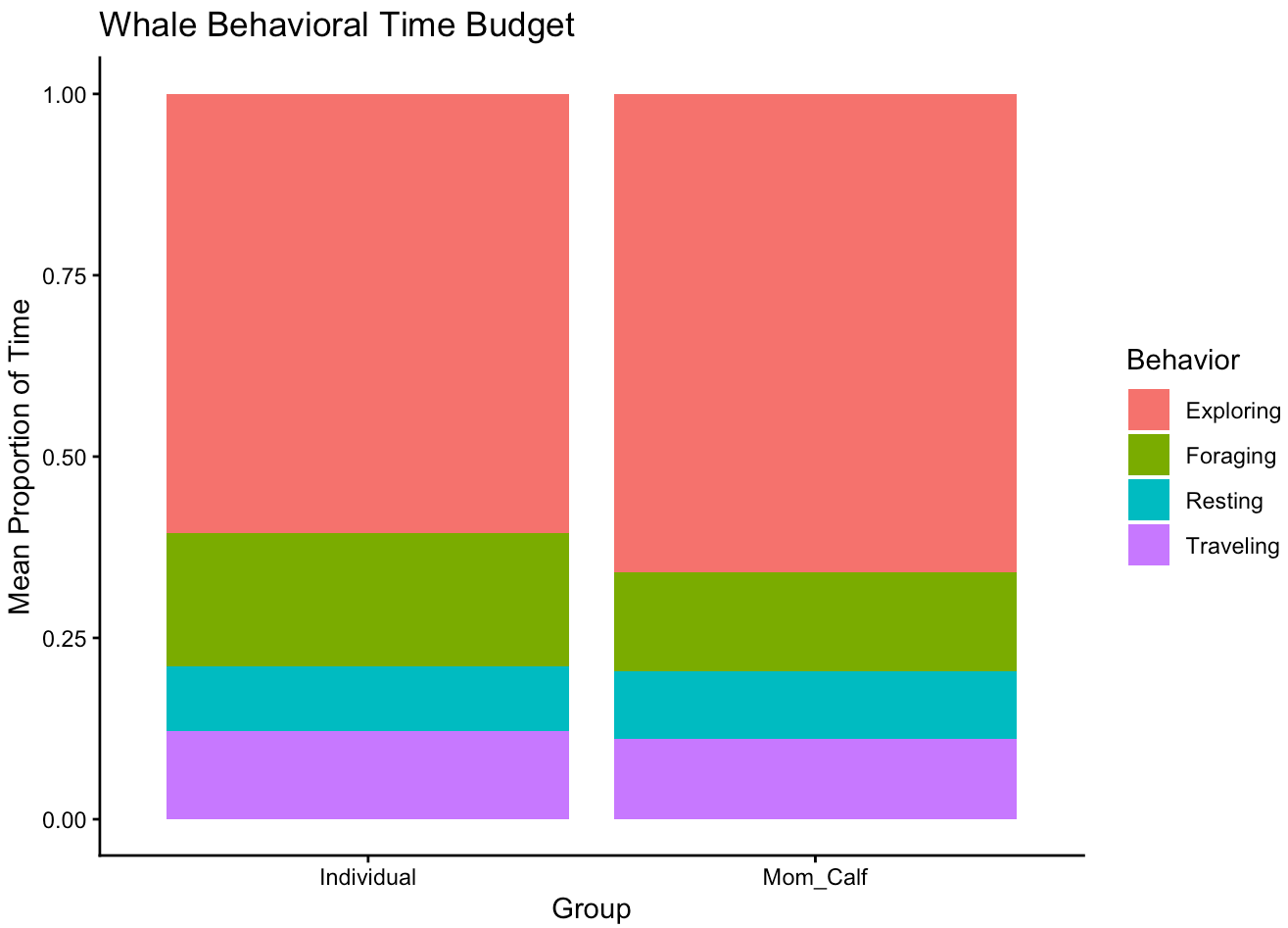



```
# ----- STACKED BAR CHART -----

behavior_summary <- Whale_Data_Filtered |>
  group_by(Group) |>
  summarise(
    Resting = mean(Resting_p),
    Traveling = mean(Traveling_p),
    Foraging = mean(Foraging_p),
    Exploring = mean(Exploring_p)
  )
```

```r
behavior_long <- behavior_summary |>
  pivot_longer(
    cols = Resting:Exploring,
    names_to = "Behavior",
    values_to = "Proportion"
  )

ggplot(behavior_long, aes(Group, Proportion, fill = Behavior)) +
  geom_bar(stat="identity") +
  theme_classic() +
  labs(
    x="Group",
    y="Mean Proportion of Time",
    title="Whale Behavioral Time Budget"
  )
```



Interpretation of Results

Beta regression models were used to test whether whale behavioral time budgets differed between **mom–calf pairs and individual whales**. For each behavior (resting, traveling, foraging, and exploring), the proportion of time spent in that behavior was modeled as a function of group. These models were fit

using **brms**, which estimates parameters using Bayesian inference and produces **95% credible intervals** for each effect.

Across all behaviors, the estimated effect of group (mom–calf vs individual) was small and the 95% credible intervals overlapped zero. This indicates that there was **no clear evidence that behavioral time budgets differed between mom–calf pairs and individuals**.

For **resting behavior**, the estimated effect of group was very small (estimate ≈ 0.007), with a wide credible interval spanning both positive and negative values. This suggests that mom–calf pairs and individuals spent **similar proportions of time resting**, and there is no strong evidence that one group rests more than the other.

For **traveling behavior**, the estimated group effect was also very close to zero (estimate ≈ –0.009), and the credible interval included zero. This indicates **no detectable difference in traveling behavior** between mom–calf pairs and individuals.

For **foraging behavior**, the estimate was negative (≈ –0.59), which may suggest a tendency for individual whales to spend more time foraging than mom–calf pairs. However, the credible interval still overlapped zero, meaning that the data **do not provide strong statistical support for this difference**.

For **exploring behavior**, the estimated effect of group was positive (≈ 0.22), suggesting that mom–calf pairs may spend slightly more time exploring. However, the credible interval again crossed zero, indicating **no clear evidence of a meaningful difference between groups**.

Overall, the analysis suggests that **behavioral time budgets were broadly similar between mom–calf pairs and individual whales**. Although there were slight tendencies for individuals to forage more and for mom–calf pairs to explore slightly more, the uncertainty around these estimates was large and the credible intervals overlapped zero. Therefore, the results indicate that **group composition did not have a statistically supported effect on the proportion of time whales spent resting, traveling, foraging, or exploring in this dataset**.

Visualizations of the behavioral data (boxplots and stacked time-budget bar charts) supported these statistical results, as the distributions of behavior proportions for mom–calf pairs and individuals overlapped substantially across all behaviors.